DAY 2 · KEYBOARD-FIRST · JULY 2026

Open-Source LLMs

Pick one. Run one. On YOUR machine. Today.

⌨️ **BEFORE WE START — open a terminal and type:**

```
ollama --version
```

Error? Raise your hand NOW — not in 20 minutes.

A model is a file. Go look at one.

⌨ DO NOW · LOOK ONLY — do NOT download! huggingface.co → search "Qwen3 GGUF" → click Files. 2 min.

WHAT YOU ARE LOOKING AT

· Hugging Face = GitHub for models.

· Each row = ONE file you could download.

· qwen3-7b ... Q4_K_M.gguf ≈ 4.7 GB

· That file IS the model. All of it.

· Inside: ~7 BILLION numbers ("weights") the model learned in training.

· We only LOOK here. Ollama does the downloading — next slide.

YESTERDAY vs TODAY

· Yesterday: model = a website. You typed into someone else's computer.

· Today: model = a file. You download it. It is YOURS.

· "Open weights" = the file is published. Anyone can run it, forever.

· "Closed" = the file stays on their servers. You rent access.

THE ONE SENTENCE

An LLM is a very large file of numbers plus a small program that runs it. Everything today follows from that sentence.

Start your download — then we talk

⌨ DO NOW · Type this now. It downloads in the background while I explain things.

```
ollama pull qwen3:0.6b
# ~520 MB - fine on any wifi, any laptop
```

Bigger laptop? Also start:

```
ollama pull mistral:7b   # ~4.4 GB, 16GB RAM
```

MEANWHILE — WHAT DID YOU JUST RUN?

· Ollama = the small program that RUNS model files.

· pull = download the file from a registry (like docker pull, npm install).

· qwen3:0.6b = family qwen3, size 0.5 BILLION weights.

· When it finishes, your laptop can answer prompts with zero internet.

WHILE IT DOWNLOADS

Keep the terminal visible. Watching the progress bar IS the lesson: that bar is 500 million numbers arriving at your laptop.

What IS a "weight"?

While your download runs: you just saw "7 billion weights" in a file listing — here is what one of them is.

A weight = ONE learned number. e.g. 0.0273 or -1.84

It controls how strongly one signal influences another inside the model. Big = matters a lot. Near zero = ignored. "7B model" = a web of 7 billion of these.

NOBODY WRITES THEM — TRAINING FINDS THEM

1. Start with 7B RANDOM numbers. Model is useless.

2. Show it "The cat sat on the ___" — it guesses wrong.

3. Nudge the numbers that would have made "mat" likelier.

4. Repeat with the next sentence. TRILLIONS of times.

Everything it "knows" is just where the numbers landed.

THE TUNING-PEG PICTURE

· An instrument with 7B tuning pegs, all random = noise.

· Training = a robot tuner nudging pegs, trillions of times.

· The GGUF file = the final peg positions. Nothing else.

· Quantization = rounding each peg slightly. 7B pegs voting together — the music barely changes.

WHY THIS EXPLAINS EVERYTHING

File size = count x bytes per number. No fact table inside — knowledge is smeared across the numbers. That is why: if you don't tell it, it doesn't know.

Your first local token

⌨️ **DO NOW · Download done? Type this. Ask it anything.**

```
ollama run qwen3:0.6b
>>> why is the sky blue?
```

WHAT JUST HAPPENED

· No internet used. Try airplane mode — it still works.

· No API key. No account. No cost. No rate limit.

· Your CPU is doing the token-by-token prediction from Day 1.

90 SECONDS OF PLAY — TRY:

· A Day 1 six-ingredient prompt. Does it obey?

· Ask something AFTER its training cutoff. Watch it guess (Day 1 lesson, live).

· Type /bye to exit when done.

· Notice the SPEED. That is 0.6B params on a laptop CPU.

MINUTE 20 MILESTONE

Every one of you is now running a language model with zero cloud. That capability is permanent. Nobody can meter it.

8 weeks. Everything changed.

JUN — Six Chinese frontier releases in TWO WEEKS. GLM-5.2 becomes the #1 open-weights model (MIT).

APR-JUN — Meta — who STARTED open weights — shelves Behemoth, delays Llama 5 to 2027, goes closed.

JUL 16 — Kimi K3: 2.8 TRILLION params. Largest open model ever. Beats Claude Opus 4.8 on some tests.

THE TORCH CHANGED HANDS · Chinese labs now hold 5 of the top-10 open slots — record high. The founder left; new stewards ship faster than anyone ever has.

WHY WE TEACH A FRAMEWORK, NOT A LEADERBOARD

Any model table is stale in 8 weeks. The six picking questions never are. Framework = the lesson. Leaderboard = homework.

One term before the map: Mixture of Experts

You will see numbers like "753B/40B" on the next slide. Here is how to read them.

753B /40B means:

753 billion weights TOTAL in the file · but only 40 billion WAKE UP per token.
The model is a building full of specialists; each question only visits a few offices.

SPEED & COST

follow ACTIVE (the small number)

DOWNLOAD & RAM

follow TOTAL (the big number)

MARKETING

always quotes TOTAL. Now you know better.

HOSPITAL ANALOGY

Kimi K3: 896 specialists in the building, your question sees 16 of them. Huge hospital, small bill.

The map · July 2026 · seven names to recognize

Family	Flagship	License	In one sentence
GLM	GLM-5.2 · 753B/40B	MIT ✓	The #1 open model in the world right now.
Kimi	K3 · 2.8T/~50B	open ✓	The LARGEST open model ever. Great at frontend.
DeepSeek	V4.1 · Flash open	MIT ✓	The reasoning + math + code specialist.
Qwen	3.5/3.6 open · 3.7 closed	Apache ✓	The multilingual all-rounder. Most downloaded.
Llama	Llama 4 (Apr 2025)	Llama lic. ⚠	The old king. Stable, huge ecosystem, no longer frontier.
Gemma	Gemma 4 · 5 sizes	Apache ✓	The laptop family. Runs on YOUR machine today.
MiniMax	M3	open ✓	The agentic multimodal newcomer.

✓ = do-anything license. ⚠ = read before commercial use. That column outlives every benchmark.

YOU DO NOT NEED TO MEMORIZE THIS

You need to RECOGNIZE the names and know the license column exists. The picking framework does the rest.

Picking a model · six questions

1 · TASK

Chat, code, classify, multilingual, reasoning?

2 · LICENSE

MIT / Apache = do anything. Others: READ FIRST.

3 · HARDWARE

Your laptop decides your size class. Be honest.

4 · LATENCY

Chat UI needs fast. Overnight batch does not.

5 · QUALITY BAR

Must it beat the frontier? Usually NO. Good-enough + private + free wins.

6 · HOSTING

Your laptop · your server · or a hosted provider.

LIVE DRILL — WE DO ONE TOGETHER

Scenario on the board: internal HR chatbot, sensitive data, 200 users, small budget. Walk the six OUT LOUD with me.

#1 · Cross-model bake-off · 20 min

⌨ DO NOW · Hosted provider (Together / Groq / HF) → 3 models, 3 families → same prompt → rank them.

PICK 3

GLM-5.2 · DeepSeek V4.1 Flash · a Qwen 3.5 — or any 3 families you like.

RUN

Identical prompt on all three:
"Explain how MoE works in 3 bullets, plain English."
Note latency, quality, length.

SCORE

Rank 1-2-3 for THIS task.
Post ranking + one surprise in chat.

BET FIRST · write your predicted winner on the card. Upset rate ≈ 50%. Loser buys coffee.

THE LESSON

Benchmarks are averages. Your task is specific. A 20-minute bake-off beats 2 hours of leaderboard reading.

Match the model to YOUR machine

⌨ DO NOW · Find your row. Write your pick on the card. You pull it in the next lab.

Why models fit at all: quantization — weights compressed to 4-bit ("Q4"). ~4x smaller, barely dumber. GGUF is the file format.

Your laptop	Runs (at Q4)	Pull this in the lab
8 GB RAM, no GPU	up to ~3B	qwen3:0.6b (already have it!)
16 GB RAM	7-8B sweet spot	mistral:7b · llama3.1:8b · qwen3:7b
16 GB + gaming GPU	up to ~13B	7B at higher quality · 13B Q4
Apple Silicon 16+ GB	13B-34B	punches above its weight (unified memory)

Small laptop = zero shame. qwen3:0.6b does the whole lab. What 0.6B does in 2026 is astonishing.

RULE OF THUMB

7B at Q4 ≈ 4-5 GB of RAM. Match model to machine BEFORE falling in love with a benchmark.

#2 · Python + your local model · 25 min

⌨️ **DO NOW** · **Ollama is already working (you proved it at minute 20). Now drive it from Python.**

```
import requests

def ask(prompt, model="qwen3:0.6b",
        system=None):
  msgs = ([{"role":"system",
            "content":system}]
          if system else []) \
       + [{"role":"user","content":prompt}]
  r = requests.post(
    "http://localhost:11434/v1/chat/completions",
    json={"model":model,"messages":msgs},
    timeout=120)
  return r.json()["choices"][0]\
       ["message"]["content"]
```

LADDER — CLIMB AS FAR AS YOU GET

1. Run the script. First Python-driven local reply.
2. Add a SYSTEM message — your Day 1 six-ingredient frame.
3. Force JSON out (Day 1 skill, local model).
4. Same prompt → your hosted provider. Compare latency + quality.
5. Swap the model name. Same code. Zero changes. (That is the point.)

WHY THE SAME SHAPE

Port 11434 speaks the same OpenAI dialect as the clouds. Learn the shape once — every provider, local or hosted, answers to it.

Debrief · what did your laptop do?

LATENCY

Local vs hosted — who was faster for the FIRST token? For the whole answer?

QUALITY

Did 0.6B obey your six-ingredient prompt? Where did it crack?

JSON

Who got valid JSON from a local model? What did it take?

SWAP

Who ran two different models with the same code? How long did the swap take?

THE HONEST TRADE

Local: private, free, offline, yours. Hosted: stronger, faster, zero ops. Pros use BOTH — the six questions tell you when.

Where this goes · 3-minute preview

Models answer. AGENTS act — they call tools, loop, and work multi-step. That is the rest of your bootcamp week. Names to recognize when you see them:

MCP	The wiring standard: any model ↔ any tool. USB-C for AI. You BUILD a server this week.
LangGraph	Agent workflows as graphs — steps, retries, approval gates. The production choice.
CrewAI	Agents as a TEAM with roles — researcher, writer, reviewer. The easy start.
Pydantic AI	Type-safe agents, JSON-first — your Day 1 structured-output skill, weaponized.

NO HOMEWORK ON THIS

Recognize the names. MCP day teaches the rest hands-on — with the local model you built today as your engine.

Cheat sheet · pin it

PICKING · 6 QUESTIONS

1. Task — what job?
2. License — MIT/Apache = free to use. Read others.
3. Hardware — your RAM decides your size class.
4. Latency — interactive or batch?
5. Quality bar — beat frontier? Usually unnecessary.
6. Hosting — laptop / server / hosted provider.

LOCAL QUICK REFERENCE

· `ollama pull <model>` — download

· `ollama run <model>` — chat in terminal

· `ollama serve` — API on localhost:11434

· Same OpenAI shape as every cloud API

· 7B at Q4 ≈ 4-5 GB RAM

· Small laptop → qwen3:0.6b, no shame

NEXT UP

HOMEWORK: the handout pack continues the ladder — chat with memory, thinking mode, your first agent. Then MCP day.

Your laptop can think now.

You proved it at minute 20 — and nobody can meter it, rate-limit it,
or take it away. The model table goes stale by September. That capability never does.

Cheat sheet · setup checklist · notebook — in your handout pack.